

ASSIGNMENT 5 (GROUP_41)

1: Implement the nonlinear model in Simulink! Use the Simulink skeleton model A5_model.slx and implement the model in the block named Vehicle model.

- Equations are

Equation for vehicle's position in Y direction

$$dY = V_x \sin(\psi) + v_y \cos(\psi);$$

Equation for Lateral velocity

$$d_vy = \frac{-(c_1+c_2)}{(m \cdot V_x)} v_y + \left(\frac{(-a \cdot c_1 + b \cdot c_2)}{(m \cdot V_x)} - V_x \right) \text{yaw_rate} + \frac{c_1}{m} \delta;$$

Equation for Heading Angle

$$d_\psi = \text{integral}(\text{yaw_rate});$$

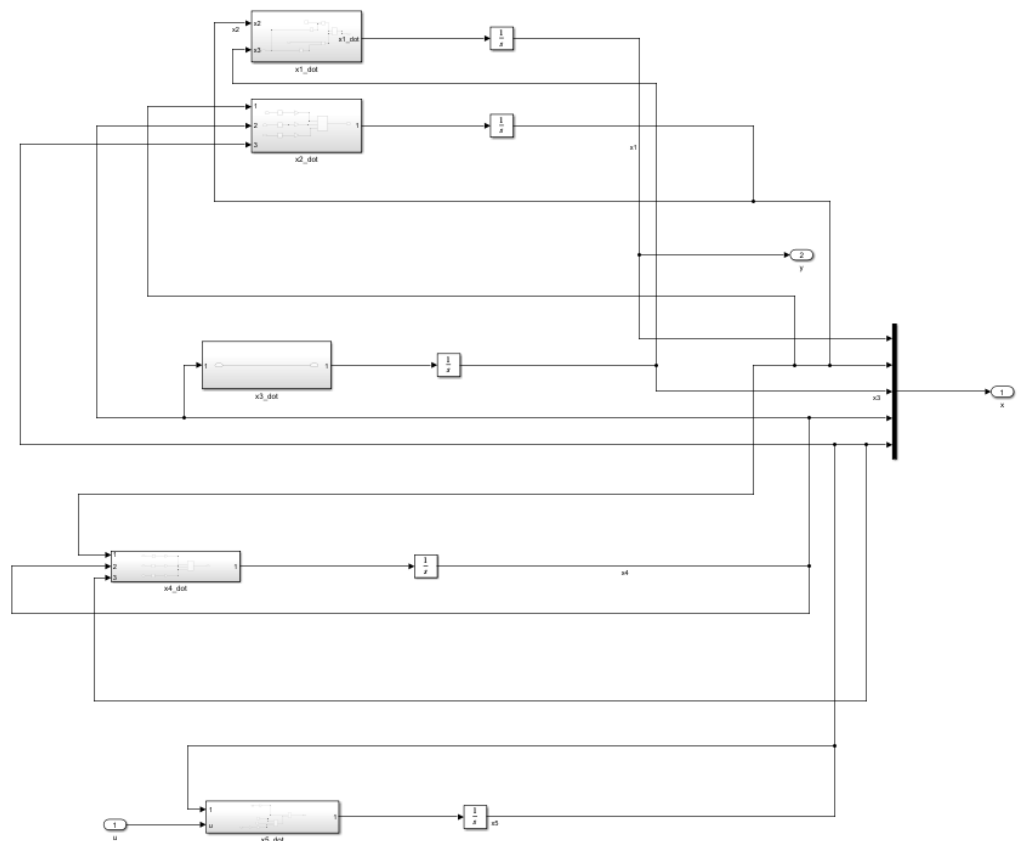
Equation for Yaw Rate

$$\% d_Yaw_rate = \frac{-(a \cdot c_1 - b \cdot c_2)}{(J \cdot V_x)} v_y + \frac{-(a^2 \cdot c_1 + b^2 \cdot c_2)}{(J \cdot V_x)} \text{yaw_rate} + \frac{(a \cdot c_1)}{J} \delta;$$

Equation for Steering Angle

$$d_\delta = -\frac{1}{\tau} (\delta - \frac{1}{2} \arctan(\frac{k_1}{V_x} u));$$

Simulink Vehicle model: (Refer 'A5_model_1')



2. Linearize the nonlinear model around the operating point, $x_e = (0, 0, 0, 0, 0)$ and $u_e = 0$ in other words straight forward driving! Answer with the A and B matrices!

Problem 2 :- State variables.

$$x = [x_1, x_2, x_3, x_4, x_5]^T = [\psi, v_y, \dot{\psi}, \dot{\psi}, \delta]^T$$

Non-Linear Equation is

$$\dot{x}_1 = v_x \sin \psi + v_y \cos \psi$$

$$\dot{x}_2 = \left(\frac{-c_1 + c_2}{m v_x} \right) v_y + \left(\frac{-a c_1 + b c_2 - v_x}{m v_x} \right) \dot{\psi} + \left(\frac{c_1}{m} \right) \delta$$

$$\dot{x}_3 = \dot{\psi}$$

$$\dot{x}_4 = \left(\frac{-a c_1 - b c_2}{J v_x} \right) v_y + \left(\frac{-a^2 c_1 + b^2 c_2}{J v_x} \right) \dot{\psi} + \left(\frac{a c_1}{J} \right) \delta$$

$$\dot{x}_5 = -\frac{1}{J} \left(\delta - \frac{1}{2} \arctan \left(\frac{k_1}{v_x} u \right) \right)$$

Linearize the non-linear around equilibrium point

$$x_e = (0, 0, 0, 0, 0) \text{ \& } u_e = 0$$

$$\dot{x} = Ax + Bu$$

Matrix A =

① Equation $\dot{x}_1 = v_x \sin \psi + v_y \cos \psi$

$$\frac{\partial \dot{x}_1}{\partial v_y} = v_x \cos(0) = 1$$

$$\frac{\partial \dot{x}_1}{\partial \psi} = v_x \cos(0) = v_x$$

② Equation $\dot{x}_2 = \left(\frac{-c_1 + c_2}{m v_x} \right) v_y + \left(\frac{-a c_1 + b c_2 - v_x}{m v_x} \right) \dot{\psi} + \left(\frac{c_1}{m} \right) \delta$

$$\frac{\partial \dot{x}_2}{\partial v_y} = \left(\frac{-c_1 + c_2}{m v_x} \right)$$

$$\frac{\partial \dot{x}_2}{\partial \psi} = \frac{-a c_1 + b c_2 - v_x}{m v_x}$$

$$\frac{\partial \dot{x}_2}{\partial \delta} = \frac{c_1}{m}$$

③ Equation $\dot{x}_3 = \dot{\psi}$

$$\frac{\partial \dot{x}_3}{\partial \psi} = 1$$

④ Equation $\dot{x}_4 = \left(\frac{-a c_1 - b c_2}{J v_x} \right) v_y + \left(\frac{-a^2 c_1 + b^2 c_2}{J v_x} \right) \dot{\psi} + \left(\frac{a c_1}{J} \right) \delta$

$$\frac{\partial \dot{x}_4}{\partial v_y} = \frac{-a c_1 - b c_2}{J v_x}$$

$$\frac{\partial \dot{x}_4}{\partial \dot{\psi}} = \frac{-a^2 c_1 + b^2 c_2}{J v_x}$$

$$\frac{\partial \dot{x}_4}{\partial \delta} = \frac{a c_1}{J}$$

⑤ Equation $\dot{x}_5 = -\frac{1}{J} \left(\delta - \frac{1}{2} \arctan\left(\frac{k_1}{v_k} - u\right) \right)$
 $\Rightarrow \frac{\partial x_5}{\partial \delta} = -\frac{1}{J}$

Matrix B
 Equation $\dot{x}_5 = -\frac{1}{J} \left(\delta - \frac{1}{2} \arctan\left(\frac{k_1}{v_k} - u\right) \right)$
 $\frac{\partial \dot{x}_5}{\partial u} = -\frac{1}{J} \left(-\frac{1}{2} \cdot \frac{d}{du} \arctan\left(\frac{k_1}{v_k} - u\right) \right)$
 $= \frac{d}{du} \arctan\left(\frac{k_1}{v_k} - u\right) \Rightarrow \frac{\frac{k_1}{v_k}}{1 + \left(\frac{k_1}{v_k} - u\right)^2}$
 $\frac{\partial \dot{x}_5}{\partial u} = \frac{k_1}{2Jv_k}$

Resulting A_lin B_lin matrices are

$$A_lin = \begin{bmatrix} 0 & 1 & v_k & 0 & 0 & 0 \\ 0 & \frac{-c_1 + c_2}{mv_k} & 0 & \frac{-a_1c_1 + b_1c_2}{mv_k} & -v_k & \frac{c_1}{m} \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & \frac{-a_1c_1 - b_1c_2}{Jv_k} & 0 & \frac{-a_1^2c_1 + b_1^2c_2}{Jv_k} & \frac{a_1c_1}{J} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{J} \end{bmatrix}$$

$$B_lin = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \frac{k_1}{2Jv_k} \end{bmatrix}$$

3. Use the linearized model derived in Problem 2. Is the system reachable?

- The solution is provided in the MATLAB.

For prob 3, A_lin and B_lin matrix are considered as A and B matrix.

The rank of the matrix is 5.

The system is Reachable.

4. Design a state feedback cruise controller, using pole placement, on the form $\Delta u = -Kx + k_r \Delta r$, such that the closed-loop system's characteristic polynomial becomes $p(s) = (s + p_1)^2(s + p_2)^3$, where p_1 (double-pole) is chosen such that specification of the lane change maneuver is expected to take 2 seconds (settling time). Select the poles p_2 (triple pole) twice as fast as p_1 . The reference gain should be chosen so the steady-state gain is 1 (use the y-position as the output signal)!

- The Solution is provided in the MATLAB. (Refer 'A5_model_1')

$P1 = 2.95$ and $P2 = 2 * P1 = 5.9$

The Poles considered are (-2.95, -2.95, -5.9, -5.9, -5.9)

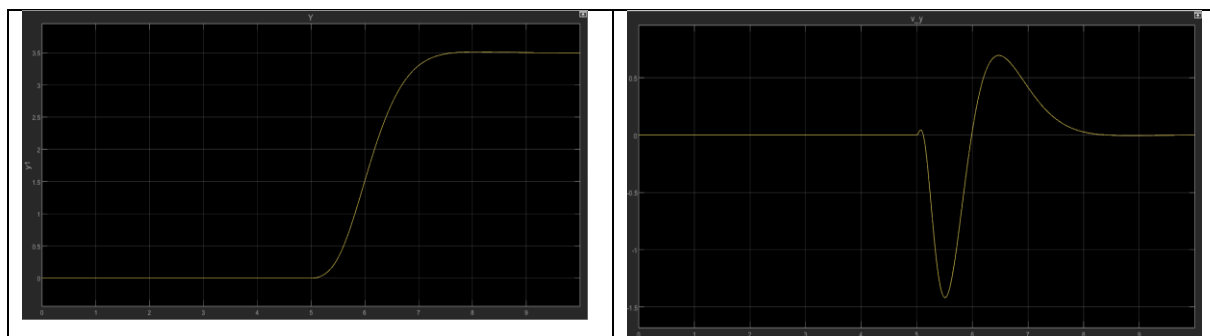
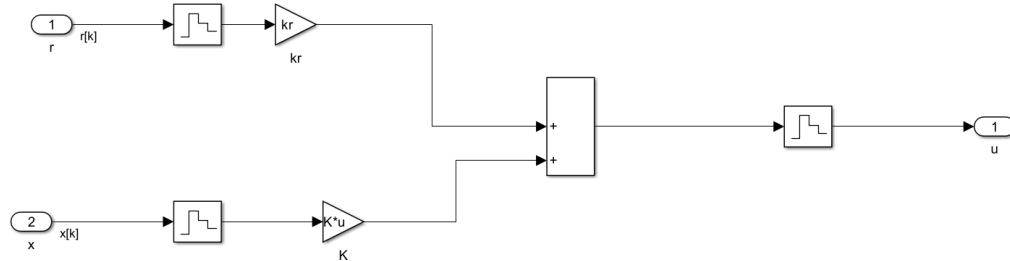
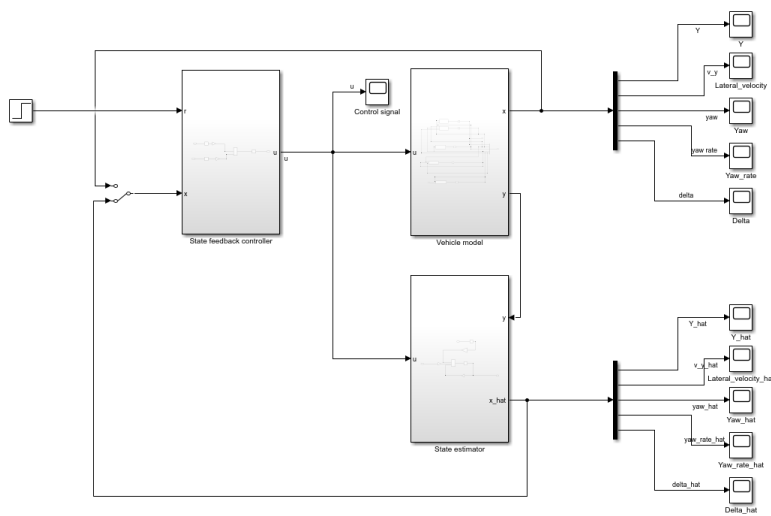
K=

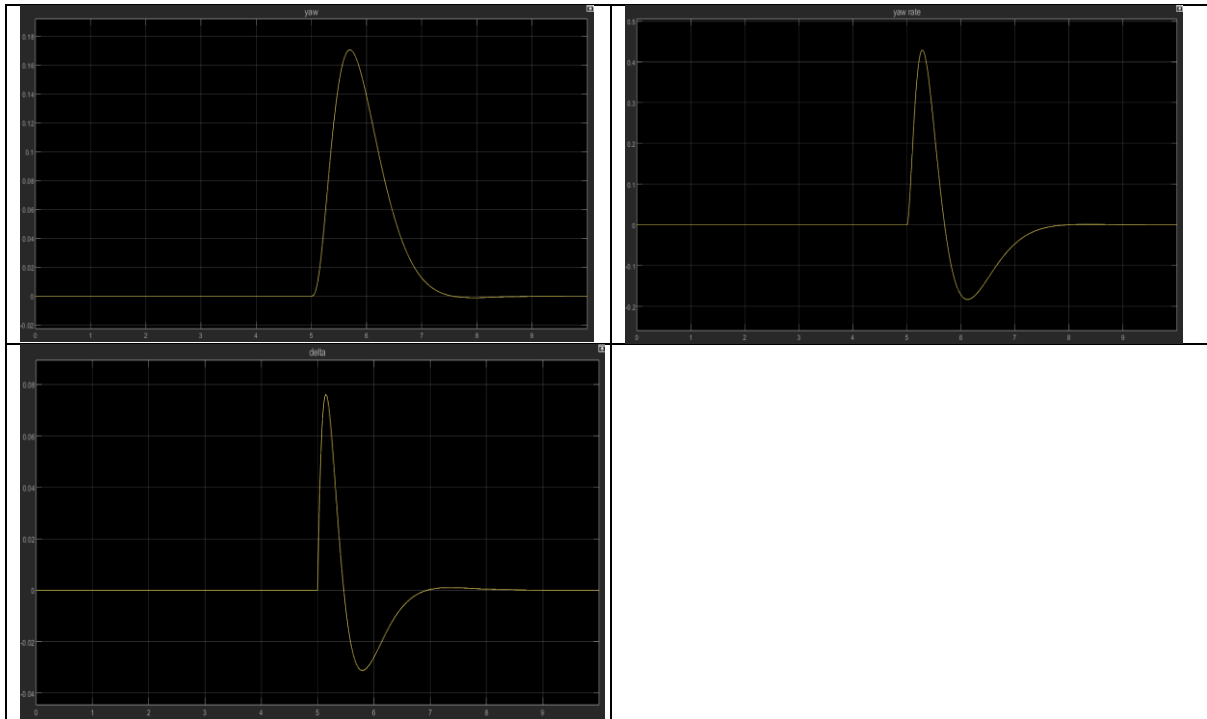
	1	2	3	4	5	6
1	0.3712	0.0156	8.4024	0.9974	1.4765	
2						

Kr =

	1
1	0.3712

Simulink model is





5. Use the concept of observability to determine which signals could be used for measurements to an observer. Is the system observable from measurements of (one or several cases might be correct), x_1 , the y position, x_2 , the lateral velocity, x_3 , the heading angle, x_4 , the yaw rate or x_5 , the steering angle? (One or several cases might be correct!)

- The Solution is provided in the MATLAB.

For prob 5, In MATLAB code A_lin matrix is considered as A .

To measure observability for Y-position

The System is Observable.

To measure observability for lateral position

The System is Not Observable.

To measure observability for Heading angle

The System is Not Observable.

To measure observability for Yaw Rate

The System is Not Observable.

To measure observability for Steering angle

The System is Not Observable.

$Obsv_vec = [1; 0; 0; 0; 0]$

6. To achieve good control performance, we place the observer poles five times faster than the feedback poles. Place all poles in the same place, in other words, quintuple poles. Where should the poles be placed?

- From the problem 4, we get the feedback desired poles as = [-2.9, -2.9, -5.8, -5.8, -5.8].

Considering the fastest pole of the feedback = 5.8.

Observer poles = 5 * P2;

$$= 5 * (5.8) = 29$$

Poles should be placed at same place = [-29, -29, -29, -29, -29]

1x5 double					
	1	2	3	4	5
1	-29	-29	-29	-29	-29
2					

7. Design a pole placement observer, which places the poles at the same place, a quintuple pole! Determine the estimator gain, L!

- The Solution is provided in the MATLAB. (For prob 7, A_lin matrix is considered as A.)

We get the eigen values as below for the determined estimator gain L.

eigenvalues

5x1 complex double	
	1
1	-29.0291 + 0.0212i
2	-29.0291 - 0.0212i
3	-28.9888 + 0.0342i
4	-28.9888 - 0.0342i
5	-28.9640 + 0.0000i

Eigen values =

L

5x1 double	
	1
1	123.2765
2	-1.9908e+04
3	1.2750e+03
4	1.1787e+04
5	386.5744

L =

8. Compute the system matrix, Ad: $x[k + 1] = A_d \hat{x}[k] + Bdu[k] + Ldy[k]$ for the discretized observer using the Euler method! Determine the estimator's poles for the discretized observer!

- The Solution is provided in the MATLAB. For prob 8, A_lin matrix is considered as A.

A_d

5x5 double					
	1	2	3	4	5
1	-0.2328	0.0100	0.2000	0	0
2	199.0789	0.9530	0	-0.2077	0.5300
3	-12.7503	0	1	0.0100	0
4	-117.8709	-0.0053	0	0.9298	0.5300
5	-3.8657	0	0	0	0.9000

A_d =

p_est_d

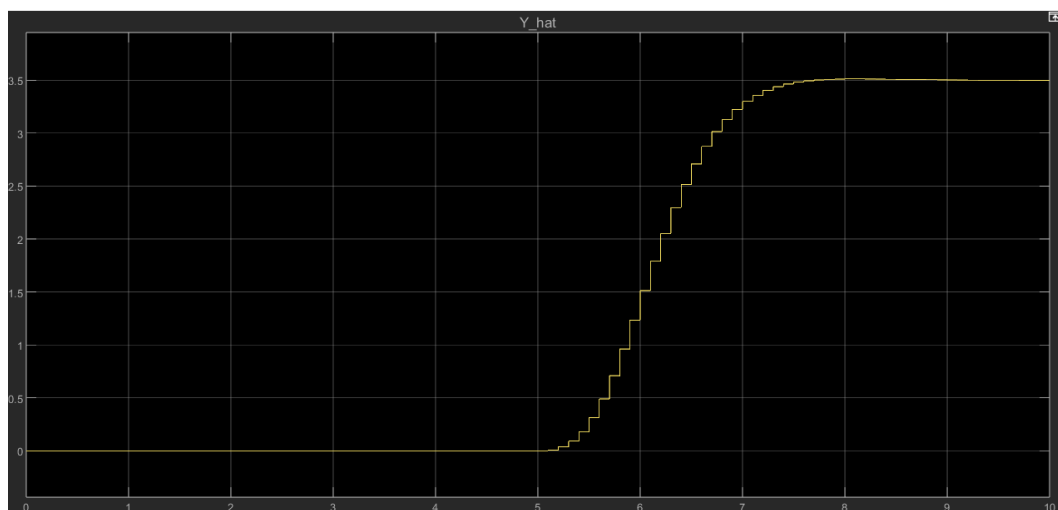
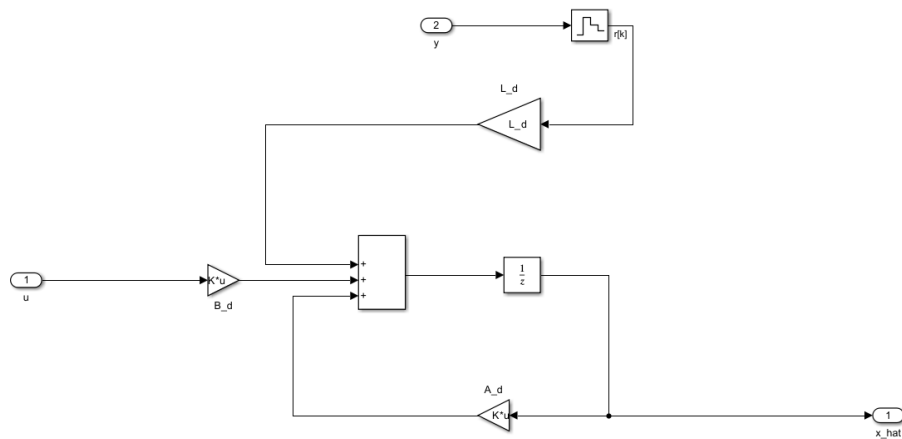
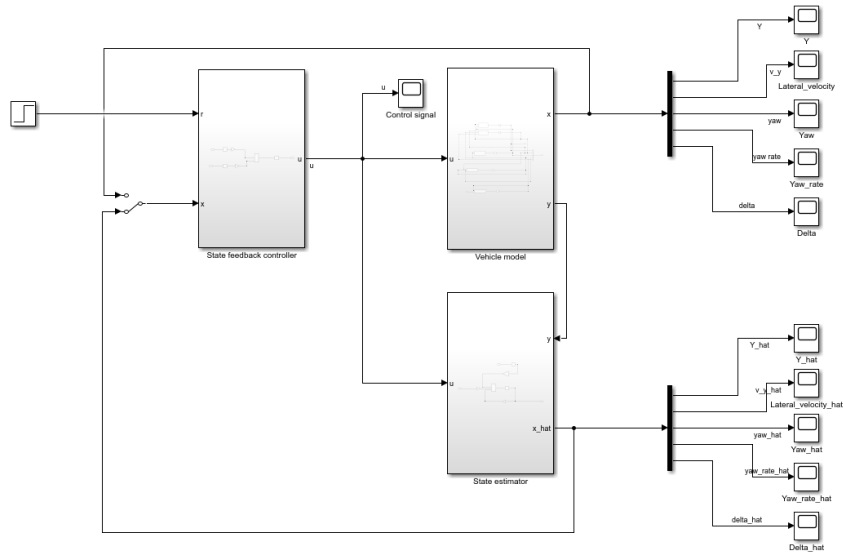
5x1 complex double	
	1
1	0.7098 + 0.0000i
2	0.7099 + 0.0002i
3	0.7099 - 0.0002i
4	0.7102 + 0.0001i
5	0.7102 - 0.0001i

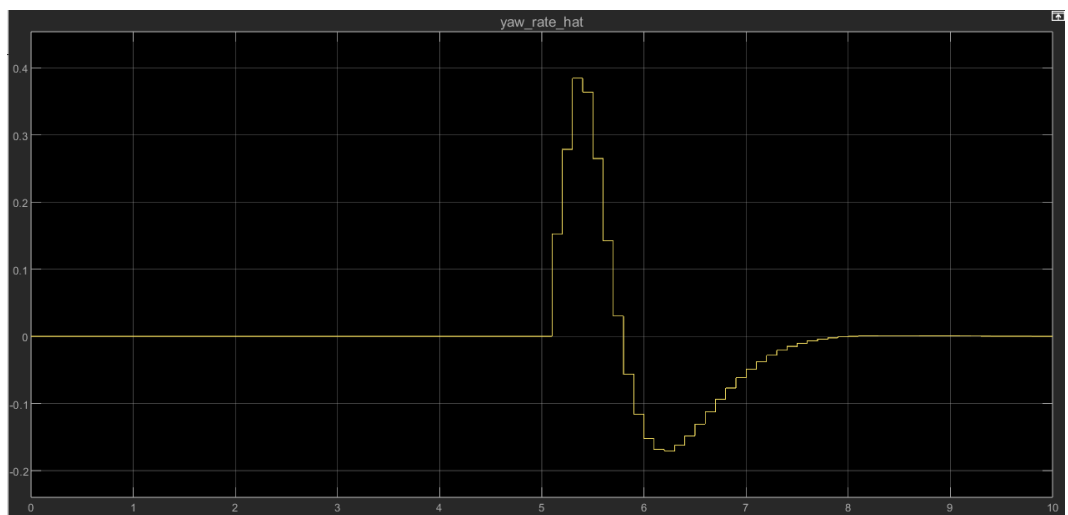
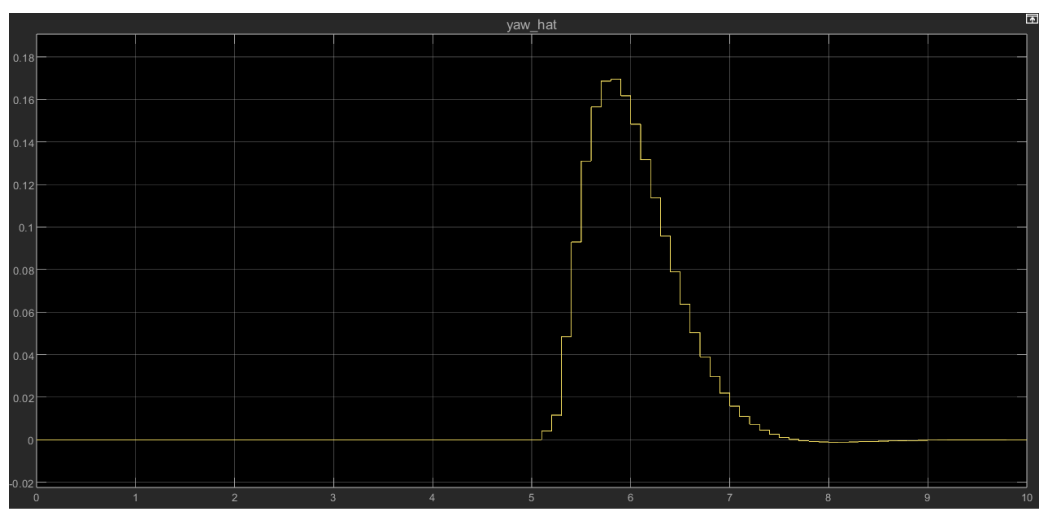
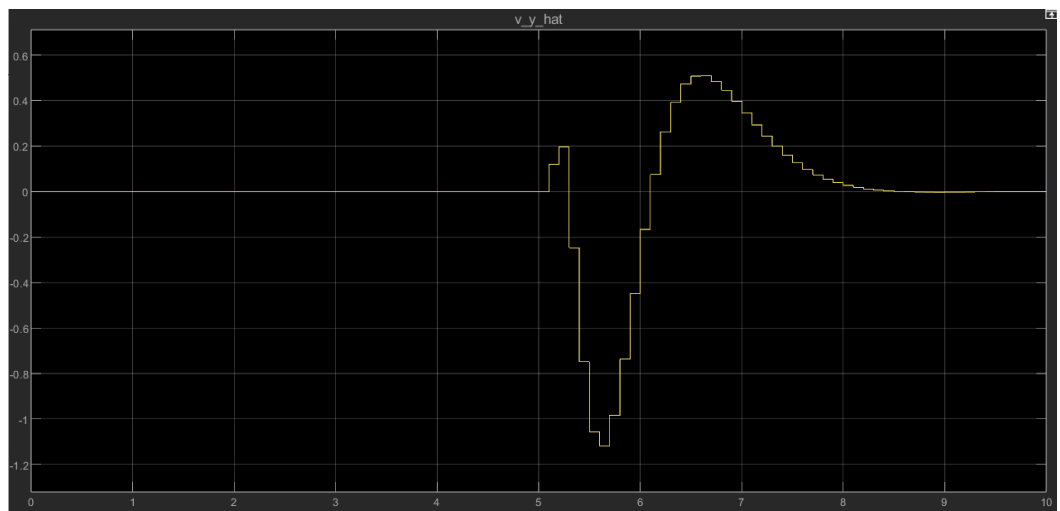
P_est_d =

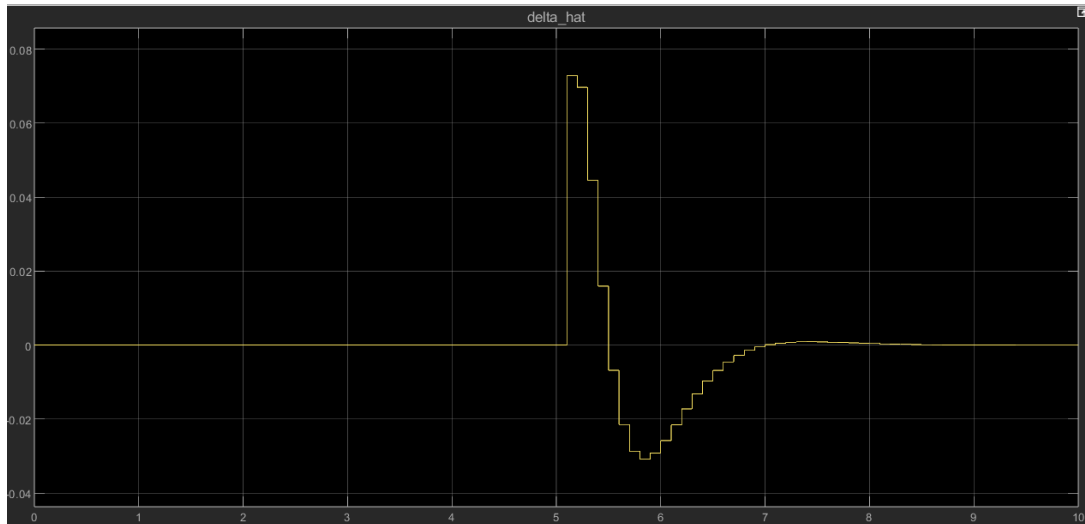
9. The last step in the design phase is implementing the discretized observer in the Simulink model! Implement the observer in the Simulink model!

- The Solution is provided in MATLAB.

Simulink Model - (Refer 'A5_model_1').







```

RiseTime: 1.2955
TransientTime: 2.0391
SettlingTime: 2.0391
SettlingMin: 0.9026
SettlingMax: 1.0035
Overshoot: 0.3497
Undershoot: 0
Peak: 1.0035
PeakTime: 3.0580

```

10. How does the controller react when we change these assumptions, i.e., what happens when we increase the mass by 400 kg corresponding to 4 passengers and some luggage and the vehicle's velocity is increased to 30 m/s? An increase in mass also increases the vehicle's inertia J with the corresponding percent change. What conclusion can be made from the design in terms of robustness? Note: You should only change the system model, not the controller or the estimator!

- The Solution is provided in MATLAB. (the values are only changed in the Vehicle model.)

Simulink Model - (Refer 'A5_model_10').

Increasing the mass by 400kg, now the mass is $m_{10} = 2400$.

Increasing the vehicle's velocity $V_{x_10} = 30$ m/s.

Due to increase in mass vehicle's inertia is increased to $J_{10} = 5220$.

```

RiseTime: 0.8898
TransientTime: 4.4725
SettlingTime: 4.4725
SettlingMin: 0.6944
SettlingMax: 1.4463
Overshoot: 20.7772
Undershoot: 0
Peak: 1.4463
PeakTime: 2.0580

```

$TF_vec = [0; 0; 1; 1]$

1. The closed loop transient performance is not affected
 - The statement is False.
2. The closed-loop system's rise time becomes slower!
 - The statement is False.
3. There is an overshoot for the closed-loop system!
 - The statement is True.
4. The closed-loop system's settling time becomes longer!
 - The statement is True.

11. If we increase the vehicle's velocity, keeping the vehicle mass and inertia as in Problem10, the system eventually becomes unstable. At which velocity (m/s) will the vehicle become unstable? Note: You should only change the system model, not the controller or the estimator!

- The Solution is provided in MATLAB. (The values are only changed in the Vehicle model.)

Keeping the mass and the inertia same as in the problem 10, $M_{10} = 2400$, $J_{10} = 5220$

At Velocity $V_{11} = 27$ m/s the system model is become unstable.